

Electric Vehicle Charge and Range Analysis using Tableau

1. Project Title

Electric Vehicle Charging and Range Analysis using Tableau Dashboard.

This project focuses on analyzing electric vehicle (EV) charging characteristics and driving range using data visualization techniques in Tableau.

2. Project Description

System Overview:

The system analyzes EV battery capacity, charging time, and vehicle driving range using interactive Tableau dashboards.

The goal is to transform raw EV data into clear visual insights.

Problem Domain:

With the increasing adoption of electric vehicles, consumers often face difficulty comparing vehicle range, battery capacity, and charging time.

This project addresses this issue by presenting EV performance data through visualization.

Core Technology Used:

Tableau for data visualization, Excel/CSV dataset for data storage, and data analytics methods for analysis.

Key Capabilities:

- EV range comparison
- Charging time visualization

- Battery capacity analysis
- Interactive dashboards

Target Users:

EV buyers, researchers, students, automobile companies, and policy makers.

Real-world Relevance:

This analysis helps users understand EV performance, supports purchasing decisions, and encourages electric mobility adoption.

3. Application Scenarios / Use Cases

Enterprise Usage:

Automobile companies can analyze EV battery efficiency and market performance.

End-user Usage:

Consumers can compare EV models based on range and charging time before buying.

Educational Usage:

Students and universities can use the dashboard to study EV performance analytics.

Industrial Usage:

Manufacturers can analyze battery performance trends to improve future EV models.

4. Technical Architecture Overview

High-Level Architecture:

EV Dataset → Data Cleaning → Tableau Visualization → Dashboard → User Interaction

Frontend Layer:

Interactive Tableau dashboards including charts, filters, and graphs.

Backend Layer:

Data processing and transformation within Tableau.

Database Layer:

Excel or CSV dataset used as a data source.

Deployment Environment:

Tableau Desktop and Tableau Public for visualization and sharing.

5. Prerequisites

Software Requirements:

- Tableau Desktop or Tableau Public
- Microsoft Excel
- Windows or Mac OS

Hardware Requirements:

- Processor: Intel i3 or above
- RAM: Minimum 4GB
- Storage: 10GB free space
- Display resolution: 1366x768 or higher

6. Prior Knowledge Required

Students implementing this project should have knowledge of:

- Basic data analytics
- Tableau interface and dashboards
- Data visualization concepts
- Dataset handling in Excel
- Basic understanding of electric vehicles

7. Project Objectives

Technical Objectives:

Develop an interactive dashboard that visualizes EV range and charging performance.

Performance Objectives:

Ensure fast visualization and accurate data representation.

Deployment Objectives:

Publish the final dashboard using Tableau Public.

Learning Outcomes:

Students will gain hands-on experience with data visualization tools and EV data analytics.

8. System Workflow

Step 1: User collects EV dataset.

Step 2: Dataset imported into Tableau.

Step 3: Data cleaning and preprocessing.

Step 4: Visualization charts created.

Step 5: Dashboard created.

Step 6: User interacts with dashboard filters and graphs.

9. Milestone 1 – Requirement Analysis and System Design

Problem Definition:

Users need a simple way to compare EV performance metrics.

Functional Requirements:

- Import EV dataset
- Analyze range and charging time
- Visualize results in dashboard

Non-functional Requirements:

- Easy usability
- Fast dashboard performance
- Clear visualization

Technology Stack:

Visualization: Tableau

Dataset: Excel/CSV

Analysis: Tableau calculations

10. Milestone 2 – Environment Setup

Steps:

1. Install Tableau Desktop.

2. Prepare EV dataset in Excel or CSV format.
3. Connect Tableau to dataset.
4. Create project folder and save Tableau workbook.

11. Milestone 3 – Core System Development

Feature 1 – EV Range Analysis:

Bar chart comparing maximum driving range of EV models.

Feature 2 – Charging Time Analysis:

Line or bar chart comparing charging time.

Feature 3 – Battery Capacity Analysis:

Pie chart or distribution visualization of battery capacity.

12. Milestone 4 – Integration and Optimization

All visual components are integrated into a single interactive dashboard.

Performance is optimized by reducing unnecessary fields and improving calculations.

Data validation is applied to ensure correct visualization.

13. Milestone 5 – Testing and Validation

Test Case Example:

Input: EV dataset

Expected Output: Dashboard visualization

Status: Pass

Performance Metrics:

Dashboard load time under 3 seconds.

Accurate visualization of EV metrics.

Smooth user interaction.

14. Deployment

Deployment Architecture:

Dataset → Tableau Dashboard → Tableau Public

Steps:

1. Create dashboard in Tableau Desktop.
2. Save workbook.
3. Publish dashboard to Tableau Public.
4. Share public dashboard link.

15. Project Structure

project_root/

├ dataset/

├ tableau_files/

├ screenshots/

├ report/

└ documentation/

Each folder stores dataset files, Tableau workbooks, screenshots, and report documentation.

16. Results

The project successfully visualizes EV performance metrics through an interactive dashboard.

Users can easily compare EV models based on range, charging time, and battery capacity.

The visualization helps simplify complex EV data for better understanding.

17. Advantages and Limitations

Advantages:

- Easy EV comparison
- Interactive visualization
- Supports data-driven decisions

Limitations:

- Depends on dataset accuracy
- No real-time EV data integration

18. Future Enhancements

- Integration with real-time EV charging station data
- Mobile dashboard applications
- AI-based EV performance prediction
- Cloud-based EV analytics platforms

19. Conclusion

The Electric Vehicle Charge and Range Analysis project demonstrates the importance of data visualization in understanding EV performance.

Using Tableau dashboards, the project converts raw EV data into meaningful insights that support better decision-making in electric mobility.